BD1_18-02-23

AUTHOR (Name, Nationality)
PUBLISHER (Name, City, Nation)
BOOK (Title, Publisher, PageNumber, Language, Price, Currency, Year)
AUTHOR_RIGHTS (Author, Book, Publisher, Percentage)

1. Define the schema of AUTHOR_RIGHTS, including the referential integrity constraints.

2. For each publisher, find the authors who have published books with that publisher in at least 10 different languages. The result should list the publishers in alphabetical order, and for each publisher, the authors should be sorted by the number of different languages (from highest to lowest).

3. Find authors who have written books for every publisher in Milan without ever receiving more than a 5% commission from them.

4. Express in optimized relational algebra and in Datalog the following query:
Find Italian authors who wrote only books in Italian before 2015 and only books in other languages from 2015 onward.

BD1_06-06-24

The following DB schema describes the Formula 1 car races. Note that ChequeredFlag is True if the pilot completes the grand prix, and that TyreType includes various predefined types (Red, Yellow, White, ...).

PILOT (PilotName, TeamName)
GRAND_PRIX (GrandPrixName, Country, Continent)
RANKINGS (GrandPrixName, PilotName, QualifyingPosition, RacePosition, Points, ChequeredFlag, TotalNrOfLaps)
USED_TYRES (GrandPrixName, PilotName, TyreType, LapsCount)

1. Define the schema of the RANKINGS table, including referential integrity constraints and attribute nullability. Express the constraint that pilots who did not finish the race must always be assigned 0 points.

2. Extract the pilots who finished a Grand Prix using three different types of tyres.

3. Extract the pilots who finished in the top three in all the Grand Prix held in Europe.

4. Build the Constructors' Standings: write a view that returns the name of the team followed by the sum of the points obtained by its pilots in descending order of points. Then find the difference in points between the first and the last team in the ranking.

5. Express in Relational Algebra and Datalog the following query:

Extract the pilots who have never finished in the top ten in European Grands Prix but arrived at least once in the top three in qualifying (considering only European Grands Prix).